# LLM private pricing channels raise the price level, with no detected contemporaneous co-movement

Apart AI Collusion Sprint · Q1.4 · checkpoint, Phase 2

**Question.** Can prices and public market state reveal colluding sellers without accusing firms that share a shock or policy?

**Finding.** Channel prices rise in **9/9 sets (7 seller models)**. Market-level markup beats pCMI in **9/9**; pCMI is not above chance in **8/9**.

**Scope.** These scores distinguish channel assignment. A price rise alone does not verify an agreement; channel-free markets can also sustain high prices.

MI measures shared information; pCMI measures residual dependence after public state and all sellers' lagged prices. A level screen measures markup over the competitive Nash price.

Source: REPORT main claim; P2.13; `claims_denominators.json`.

# Validation: opposite failure modes

Four sellers, repeated differentiated-price market. Scripted sellers validate detectors; LLM sellers test the substantive claim. Frozen estimators, predeclared Phase 2 analyses.

| scripted condition | level-screen flags | pCMI flags |
|---|---|---|
| planted shared-signal coordination | see REPORT §4.1 | 100% |
| unequal private costs, no channel | 30/30 | 4/30 |

**Conditioning helps but does not prove intent:** raw MI flags 100% of common-shock markets; pCMI flags 14% under the shift null (target 5%). Shared random seeds and legitimate leader-follower pricing remain confounds.

Sources: `gate_test.json`, `results/p2/costasym_test.json`; REPORT §4.1, P2.4. Flag counts use different registered nulls in these two experiments.

# Across 9 sets: channel assignment is mostly a level

| seller | market-level AUROC [95% CI] | pCMI AUROC [95% CI] |
|---|---|---|
| luna (T=20) | 0.94 [0.86, 1.00] | 0.47 [0.28, 0.67] |
| deepseek (T=20) | 0.99 [0.96, 1.00] | 0.57 [0.36, 0.77] |
| luna (T=60) | 0.94 [0.86, 1.00] | 0.34 [0.18, 0.52] |
| Qwen3.8 abl. (T=20) | 0.95 [0.87, 1.00] | 0.52 [0.29, 0.74] |
| Qwen3.8 base (T=20) | 0.96 [0.89, 1.00] | 0.46 [0.25, 0.68] |
| older luna (T=20) | 0.75 [0.50, 0.94] | 0.72 [0.47, 0.92] |
| Qwen3.8 abl. + memory (T=40) | 0.85 [0.71, 0.97] | 0.49 [0.30, 0.67] |
| Qwen3.6 abl. (T=20) | 0.96 [0.87, 1.00] | 0.41 [0.18, 0.63] |
| Qwen3.6 base (T=20) | 0.93 [0.81, 1.00] | 0.71 [0.51, 0.88] |

10–21 games per arm; CIs bootstrap games. The sets share designs/models. Base Qwen3.6 is the pCMI exception: 0.71 [0.51, 0.88].

Sources: `p2_partc.json`, P2.13 inclusion list. Non-channel benchmarks are model- and prompt-specific.

# Longer games and executed agreements do not rescue pCMI

| registered comparison | market-level AUROC [95% CI] | pCMI AUROC [95% CI] |
|---|---|---|
| E44, T=60, 20 + 20 games | 0.94 [0.86, 1.00] | 0.34 [0.18, 0.52] |
| E65, Qwen memory, T=40 | 0.85 [0.71, 0.97] | 0.49 [0.30, 0.67] |

**Executed only:** 48/149 channel games over 10 sets; pooled pCMI 0.42 [0.33, 0.53]. Execution uses agreement + kept floor + supra-Nash price. Selecting on high prices inflates the level screen; no such level AUROC is claimed here.

E65's declared executed-only test is **inconclusive** (5 executed); memory is not a confirmed rescue. E44's planting check establishes sensitivity to one shared-move alternative, not every form of collusion.

Sources: `long_power_llm_long_p2_T60.json`, `p2_partc.json`; REPORT P2.8, P2.11.

# The reference market determines what a flag means

| context | finalized evidence |
|---|---|
| five prompt wordings | no-channel mean markup -0.19 … +2.56 |
| older gpt-5.6-luna | 9/10 no-channel markets tacit-high |
| luna + memory + strong profit framing (Q2.2) | 19/20 tacit-high without a channel; 13/20 agreements executed with one |

Under strong framing the declared detector test is **not evaluable**: competitive no-channel markets are too rare. Channel labels stop being ground truth.

**Abliteration is no simple explanation.** The channel premium does not detectably increase in the Qwen pairs tested (REPORT P2.11b); serving precision is a caveat. E69 interaction, abliterated − base: −0.65 [−1.97, 0.65].

Sources: `p2_variants.json`, `p2_partc.json`, `strongF_ext.json`; REPORT P2.7, P2.11.

# Use screens to trigger an audit, not to deliver a verdict

- **Combine evidence.** Absolute levels need cost data; dependence misses sustained levels. Relative pair screens lose power when prices rise market-wide.
- **Match the reference.** Model, prompt, board availability and costs matter. At 1% collusion prevalence no tested frozen flag reaches point-estimate precision 0.25 (P2.1).
- **Validate execution separately.** Opportunity, agreement and kept pricing are different labels. High prices without a channel need their own label.
- **Generality and transfer:** registered 16-cell grid (E70, 192 games): C − A positive in 13/16 cells, level > pCMI in 12/16 (point estimates); lagged dependence (TE) weakly above chance, 0.62 [0.54, 0.70]; exploratory E72: TE names the channel pair 30/96 (chance 1/6), signal in non-executed channel games (0.73 vs executed 0.49; post hoc). External double auction (E68): inconclusive, chat raises asks only +0.39 [+0.06, +0.69]. The declared E70 follow-up (factor: model) was not confirmed: +1.07 [−0.20, +2.41].
- **Limitations:** one core market family, modest samples, unadjusted comparisons and estimator-specific blind spots. A combined screen-and-audit pipeline is a proposal, not a tested deployment.

Full evidence: `REPORT.md`; commands in §6; all experiments, corrections and duds in `EXPERIMENTS.tsv`. Markdown and figures reflect this checkpoint; PDFs retain the last scheduled export.